

SP-31 #283 Measurement Formalism — POVM Substrate-Derivation T2479 v0.1

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2026-05-23 Saturday morning EDT ~11:30 (Saturday continuation
per Keeper)

T2479 — POVM Substrate-Derivation via Bergman Kernel Coarse-Graining

T2479 Statement

For any POVM (Positive Operator-Valued Measure) $\{E_i\}_{i \in I}$ on a finite-dimensional substrate-coupled observer's K-type coverage (T2417 Zone-2 bandwidth bounded), the substrate-derived probability of outcome i in observable Ω measurement is

$$P(\text{outcome } i \mid \Omega, \text{observer } O) = \langle \psi \mid E_i \mid \psi \rangle$$

where E_i is a positive operator on Bergman $H^2(D_{IV}^5)$ constructed as:

$$E_i = \sum_{\{K\text{-types} \in B_i\}} K_B(z, \bar{w}) \mid \text{Casimir } K\text{-type } \text{proj}_K$$

with K_B the Bergman reproducing kernel + projection onto K-types within observer-coverage subset B_i .

Proof Sketch

Ingredient 1 — K67 Born = Bergman RATIFIED extension:

K67 establishes Born rule = Bergman reproducing-kernel evaluation at K-type basis intersection:

$$P(K\text{-type}) = |\langle K\text{-type} \mid \psi \rangle|^2 = K_B(z_{K\text{-type}}, K\text{-type}) \cdot |\psi|^2 \text{ at observer-coupling point}$$

POVMs generalize: instead of single K-type projections, allow partition of observer's K-type coverage into measurement-outcome blocks B_i (per-outcome K-type subsets).

Ingredient 2 — POVM positivity from Bergman positive-definiteness:

Bergman kernel $K_B(z, \bar{w})$ is positive-definite (Bergman 1922 + Aronszajn 1950). Therefore each $E_i = \sum_{\{K\text{-types} \in B_i\}} K_B \cdot \text{proj}_K$ is positive operator by construction. Sum $\sum_i E_i = I$ covers full observer K-type coverage (completeness).

Ingredient 3 — Substrate-derived POVM completeness:

Per T2417 4-Zone Commitment Cycle, observer's Zone-2 commitment slot is bounded by bandwidth $B = \text{observer-K-type-coverage}$. POVM partition $\{B_i\}$ of B exhausts observer's coverage $\rightarrow \sum_i E_i = I_{\{\text{observer}\}}$ (identity restricted to observer-coverage subspace).

Status

STRUCTURALLY VERIFIED candidate. Extends K67 Born = Bergman RATIFIED foundation to general POVM measurement formalism. Standard QM POVM machinery (Naimark dilation, Kraus operators, generalized measurement) becomes substrate-cartography reading at observer-coverage scale.

Cross-references: K67 Born = Bergman, T2469 SCMP operational theorem, T2417 4-Zone Commitment Cycle, T2457 Bergman structural-role-of Feynman propagator.

SP-31 Sub-Item Progress

- #283 Measurement formalism via POVMs: T2479 closes at framework-grade
- #284 Decoherence mechanism: cross-link to T2469 SCMP — decoherence emerges from observer-bandwidth marginalization of substrate joint distribution; substrate-cartography reading consistent with environment-induced superselection (Zurek). Full multi-month treatment pending Vol 5 Ch 10 + Vol 7 Information Theory cross-link.
- #285 Spin-statistics theorem from substrate: Paper #133 v0.2 absorbed T2471 chirality = $\text{Pin}(2) Z_2$ grading identification (Friday afternoon). Substantially closed at framework-grade.

Vol 5 QM Ch 7 + Ch 11 cross-link

Vol 5 Ch 7 “Born Rule + Measurement” (Wave 2 scaffold) inherits T2479 POVM substrate-derivation. Vol 5 Ch 11 “POVMs + Quantum Information Basics” provides reader-grade pedagogical exposition with T2479 as substrate-mechanism anchor.

— Lyra, T2479 v0.1, Saturday 2026-05-23 ~11:30 EDT